

PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – Overview of IT Industry

MODULE - 1

SESSION / ASSIGNMENT – 1

TASKS :

- 1. Name 5 IT job roles and write one line about what each role does.**

➔ **1. Front-End Developer**

Builds the user interface of websites and applications that users see and interact with.

2. Back-End Developer

Develops the server-side logic, databases, and APIs that power applications behind the scenes.

3. Full-Stack Developer

Works on both the front-end and back-end parts of an application.

4. QA Tester

Tests software to find bugs and ensure quality before release.

5. Data Analyst

Analyzes data and creates reports to support business decisions.

- 2. List 3 product companies and 3 service companies with one example of work each does.**

➔ **Product Companies**

1. Microsoft – Develops the Windows operating system and Office products.

2. Google – Creates products such as Android and Google Search.

3. Adobe – Develops software like Photoshop and Premiere Pro.

Service Companies

1. Tata Consultancy Services – Provides software development and IT consulting services.

2. Infosys – Offers application development and digital transformation services.

3. Wipro – Provides cloud, cybersecurity, and IT support services.

3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

➔ **1. GitHub** - [**Git** (a distributed version control tool) + **Hub** (a central hosting platform)
= **GitHub** (Git-based code hosting platform)]

Used to store, manage, and collaborate on source code.

2. Slack - Searchable Log of All Communication and Knowledge

Used for team communication and collaboration.

3. Jira - Not officially an acronym

Used for project management and bug tracking.

4. IDE - Integrated Development Environment

Used to write, run, and debug code.

5. API - Application Programming Interface

Allows different software applications to communicate with each other.

4. Name any 5 programming languages and mention one popular use case for each.

➔ **1. Python** – Machine Learning

2. JavaScript – Web Development

3. Java – Android App Development

4. C++ – Game development

5. PHP – Backend Web Development

5. What is the difference between a developer and a tester? Answer in 2 lines.

➔ A **developer** writes and maintains the code that makes a software application work.

A **tester** checks application for bugs, verifies, ensures the software works correctly before release.